



Deep CP-CXR: A Deep Learning Model for Classification of Covid-19 and Pneumonia Disease Using Chest X-Ray Images

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Abstract

The global spread of the Coronavirus has caused a disastrous effect, affecting millions of people and making it crucial to take action. Numerous experts have worked extensively to create viable vaccines in the fight against this infectious disease. The current study offers hope by suggesting a deep learning model, Deep CP-CXR, for determining patients with Covid-19 and pneumonia. Our study encompasses two significant investigations. First, we used images from chest X-rays for binary classification to distinguish Covid-19-diagnosed patients from normal patients. Second, using chest X-ray images, we expanded the study to include several groups, such as pneumonia, Covid-19, and normal instances. The results of our studies were extremely promising. The binary classification achieved a remarkable average accuracy of 100%, allowing for accurate classification between Covid-19 patients and normal cases. In addition, the multiple-category classification was able to distinguish between Covid-19, pneumonia, and normal individuals with a remarkable average accuracy of 98.57%. These astounding findings lead us to conclude that the Deep CP-CXR method we've suggested for classifying Covid-19 and pneumonia patients enables medical professionals to perform it accurately. Healthcare providers worldwide will benefit significantly from this development since it has the potential to enhance both the detection and treatment of these ailments. The proposed deep learning approach improves the speed and precision in classifying the disease with which doctors can diagnose and treat their patients effectively.

Keywords Covid-19 · Pneumonia · Deep learning · Classification · Chest X-ray

1 Introduction

The 2019 coronavirus disease (COVID-2019) has been rapidly spreading over the globe since the start of 2020, with its origins in Wuhan, China. Numerous fatalities and millions of illnesses have resulted from it. The world has started to mobilize and adapt various restrictions and preventative measures to stop the spread of this illness. Identifying the sick person is also necessary to control the infection. On the other hand, because of a lack of reverse transcription-polymerase chain reaction (RT-PCR) assays, chest computed tomography (CT) has grown in favor of aiding in the diagnosis of COVID-19 [1]. According to the World Health Organization, there are 163 million recovered cases and 178 million coronavirus cases (WHO). However, there are still more than 3 million cases of death globally. The USA, Brazil, Italy, and India are the nations that are most frequently impacted by coronaviruses worldwide. One of the many viruses of the Nidovirales order, including the Roniviridae, Coronaviridae, Mesoniviridae, and Arteriviridae families, is the coronavirus. They are all enveloped unsegmented sense ribonucleic acid (RNA) viruses and belong to the order Nidovirales. They are all RNA viruses with giant genomes. The coronavirus's positive-stranded, 30 kilobase (kb) genome is not segmented. Nidovirales and Coronaviridae also rapidly spread in humans and mammals [2–4]. Different diseases, ranging from symptomatic to asymptomatic infections, can be brought on by the COVID-19 virus. Real-time PCR (RT-PCR) testing is the accepted method for identifying verified cases [5].

Vaccination is required to combat the Covid-19 virus, which is causing an epidemic over the globe. Do WHO and its employees collaborate to monitor the outbreak, direct essential initiatives, and provide basic medical supplies to needy people? The vaccination saves lives. The immune system is trained and shaped by vaccinations to recognize and eliminate the viruses and germs they target. If these hazardous germs do not later shield the body after immunization, the body can quickly destroy them, averting the sickness. On February 18, 2021, countries revealed more than seven vaccinations across three platforms. Vaccination is given the greatest priority for vulnerable populations worldwide. 60 of the 200 new candidate vaccines being developed have reached the clinical stage. COVAX is integral to the ACT Accelerator, which was launched by WHO and partners in 2020. The goal is to end this phase of the Covid-19 pandemic [6]. Pneumonia is also a life-threatening disease, usually caused by a bacterium called *Streptococcus pneumonia*, which affects the human lungs [7]. Pneumonia kills about 700,000 youngsters every single year and influences 7% of the worldwide population [8]. “Deep CP-CXR, a deep learning model, is proposed in this study to recognize the COVID-19 patient and pneumonia disease. Based on a deep neural network, convolutional Neural Network (CNN) has been utilized to extract features from the chest X-ray images. The following are the objectives and primary contributions:

- This paper introduces the Deep CP-CXR algorithm, a deep learning algorithm that accurately identifies pneumonia and COVID-19 in chest X-ray images.
- Two tests (binary-class and multi-class) evaluate the model's diagnosis techniques.

- The model achieved 100% accuracy in binary classification and 98.57% in multi-class classification, demonstrating its proficiency in illness differentiation.
- The integration of Deep CP-CXR in medical diagnostics has led to considerable improvements in efficiency and accuracy in disease classification.

Deep learning models, such as CNNs represented by Deep CP-CXR, exhibit exceptional precision in disease classification from chest X-rays by identifying critical features. This model facilitates the automatic and prompt identification of diseases such as COVID-19 and pneumonia, which is essential in urgent situations. Engineered to manage extensive datasets, it is optimal for high-demand healthcare environments. The model, trained on varied datasets, demonstrates effective generalization, guaranteeing precise classifications among distinct patients. Methods such as heatmaps improve interpretability, aiding healthcare personnel in identifying crucial regions in X-rays. The model assists radiologists by delivering preliminary screens, consequently reducing their diagnostic workload.

This research is structured as follows for the subsequent sections: Sect. 2 presents the literature review. Section 3 delineates the Research Methodology, encompassing data set description, data preprocessing, proposed classification framework, early notions, and the proposed Deep CP-CXR model. Section 4 presents the experimental evaluation and results, encompassing the implementation environment, performance metrics, parameter tuning, results, and comparative analysis. Sections 5 and 6 address the discussion and conclusion about future scope.

2 Literature Review

2.1 Deep Learning Models for COVID-19 and Pneumonia Detection

The research examined multiple diagnostic models based on traditional methods and deep learning, as per the motivation behind it. Inspired by this, we anticipated a single model for primarily detecting COVID-19 and pneumonia. Authors in [9] Introduced the research on the importance of the Zhongzhi set for deep switching to acquire fashion knowledge on the constrained Covid-19 chest X-ray data collection. This observation is based on neutrosophic set theory. The neutrosophic setting in this view is used to change the clinical picture from the gray-scale space area to the neutrosophic area. The Zhongzhi District includes three styles of photos: real (T) photos, uncertain (I) photos, and fake (F) photos. The data collection utilized in this research is accumulated from unique reiterations. The data set includes four instructions: Covid-19, normal, and pneumonia bacteria/virus. According to the experimental results, the Indeterminacy (I) neutrosophic area has reached the best possible correctness in the verification precision and overall performance index composed of Precision, F1 Score, and Recall.

A Bayesian optimization model (SqueezeNet) utilized momentum parameters and learning rate. The author used 5,949 X-ray images, comprising 1,583 standard samples, 4,290 pneumonia samples, and 76 verified Covid-19 samples. The author achieved an accuracy of 100% for Covid-19, 96.73% for pneumonia, and 98.04% for

normal patients [10]. A deep learning, reusable, and problem-based model to identify consolidation on chest X-ray images to help radiologists better diagnose pneumonia [11], with an accuracy of 94.67% for detecting consolidation. This model helps detect pneumonia, but the accuracy rate is low. Research [12] projected a DCNN to classify pneumonia employing chest X-ray images. The author in [12] suggested an enhanced DCNN model in which valuable features have been first hauled out from the pictures and then categorized into normal and pneumonia categories. The suggested method has been trained and tested on the chest X-ray image dataset.

2.2 CNN and Image Enhancement Techniques

The study [13] reviewed a multi-image enhancement technique primarily based on CNN to recognize COVID-19 in the experimental pixels of chest X-rays and chest CT of a suspected coronavirus individual. Multi-picture enhancement uses discrete records obtained in filtered pixels to increase the number of powerful examples used to train the CNN version. Using this method, the proposed well-known version showed higher class accuracy in CT experiments and X-ray pixels, respectively, with 95.38% and 98.97%. Chest CT is likewise a powerful method to accidentally identify Covid-19. Author in [14] proposed an analysis machine learning framework that employs specious X-ray images of patients to identify health, pneumonia bacterial/viral, and Covid-19. The attainment of a whole accurateness rate of 92.4% at 10-fold cross-validation.

2.3 AI-based CT Analysis for COVID-19

The authors in [15] proposed a powerful technique to recognize Covid-19 patients, primarily based on chest CT images. Authors in [16] developed an AI-driven CT tool for the analysis and measurement of COVID-19. The suggested module can autonomously extract lung slices. This model achieved a sensitivity of 98.2% and a specificity of 92.2% on the target dataset. To reduce the variety of patterns [16], adopted transformation learning to transfer the expertise extracted in pre-skill mode to the version that requires the skill. Experimental results display the overall performance of this category can reach 95% accuracy for the two fine fashions.

2.4 Transfer Learning and Hybrid Models

The authors [17] developed an enhanced integrated transfer learning technology, which significantly improves performance compared with traditional transfer learning. The architecture of each network has been altered with additional layers through a collection of deep learning models. The accuracy of this technique for binary classification is 99.49%, and the accuracy for multi-class classification is 99.24%. In addition, authors in [18], five pre-trained CNN models viz. NASNetMobile, MobileNetV2, VGG16, InceptionResNetV2, and Xception have been trained and tested. They augment the image by performing data augmentation and Generative Adversarial Networks (GAN). Among these models, VGG16 and Xception show better results than models trained with enhanced images.

2.5 COVID-19 and Vaccine-Related Studies

The Covid-19 vaccine is widely regarded as an important aspect of the pandemic recovery in the UK, and a complete distribution method may be mandatory to maximize absorption. Nevertheless, there may be a lack of investigation on the causes that may cause British residents to be attracted to or reject the Covid-19 vaccine. This has focused on using discrete preference testing to study the significance of vaccine characteristics, transportation, and media insurance in expanding/reducing vaccine intake. Effectiveness becomes the aspect that determines the greatest impact on vaccine selection; in addition, for older adults over 55 years of age, the huge impact of excessive efficacy is becoming more and more. The insights of this DCE are designed to help policymakers and public fitness communicators develop plans for the Covid-19 vaccine and improve its transportation methods [19]. For covid-19 vaccines by the US Food and Drug Administration (FDA), radiologists should keep current Covid-19 immunization records in mind for patients with the following disorders. Unilateral axillary lymph node enlargement underwent a different analysis. On ultrasound examination, hyperplasia of axillary lymph nodes can be noticed after any vaccination. Still, hyperplasia of axillary lymph nodes is more likely following immunization with the Covid-19 vaccine, which can generate a robust immunological response. Because breast cancers are included in the differential diagnosis of unilateral axillary lymphadenopathy, it is critical to compare the results. Because breast cancers are included in the differential diagnosis of unilateral axillary lymphadenopathy, it is critical to compare whether the breast is the first malignancy and to obtain updated vaccination records. Because the Covid-19 vaccine will shortly be administered to more people, radiologists ought to be at ease with the imaging capabilities of Covid-19 vaccine-induced hyperplastic adenopathy and how it can be utilized to diagnose unilateral axillary adenopathy. Instead of a rapid useless and expensive axillary lymph node biopsy, temporary continuation for unilateral axillary lymphadenopathy after receiving the new Covid-19 vaccination is the best advice [20].

2.6 COVID-19 Detection Using Chest X-ray Images

Authors in [21] proposed cost-sensitive and discriminative learning procedure for identifying Covid-19 employing 2,239 chest X-ray images incorporating 239 from confirmed COVID-19 cases, 1,000 from confirmed bacterial or viral pneumonia cases, and 1,000 from healthy individuals. The author attained an accuracy of 97.01%, a precision of 97%, a sensitivity of 97.09%, and an F1-score of 96.98%. In addition, authors in [22] suggested a DNN-based model named COV-SNET, which was pre-trained on more than 1,000 chest X-ray images. In this research [22], considered the sensitivity of two COV-SNET models to analyze COVID-19 has reached 95%. Authors in [23] Use Convolutional Neural Network (CNN) to analyze lung radiographic images to differentiate Covid-19 from noncommunicable cases and additional viral/bacterial lung disease styles. 100% of sensitivity, as well as 97.6% of specificity, has been attained. Authors in [24] proposed a deep learning-based model named “LiteCovidNet” and CoBiD-net for time series forecasting of Covid-19. The study helps countries know the severity of Covid-19 concerning education in the

future along with forecasting Covid-19 cases and human body interaction with the Covid-19.

2.7 Other Novel Techniques and Approaches

Authors in [25], deep vision (BoDVW) method is suggested by authors for discovering Covid-19 and pneumonia diseases. The researchers used 4 available data sets of Covid-19 CXR images viz. D1, D2, D3, and D4 where the accuracy of D1 was 82.00%, D2 was 87.86%, D3 was 87.92%, and D4 was 83.22%. Another recent technique is the use of MCFF-Net to sense Covid-19 in chest X-ray images [26]. The study's data set includes four types of pneumonia: Covid-19, pneumonia bacterial/viral, and normal. The experiment was 97.30% accurate on average. Furthermore, the accuracy rate is 95.02%, the recall rate is 95.27%, the specificity is 98.20%, and the f1-score is 94.79%, according to the experimental data.

3 Research Methodology

3.1 Dataset Description

The chest X-ray dataset is used to train and evaluate the proposed method. Using samples of the same type covering the assessment category is essential to ensuring the model's performance is accurately validated. Then, the same number of photographs are used for testing as well as training. There are various images and references in Table 1. 5228 samples total, comprising pneumonia, normal, and Covid-19 were utilized. Multiple data sets were infected to validate and test the model. For Covid-19, Normal, and Pneumonia, 1626, 1802, and 1800 sample images were selected to validate the proposed model. Six distinct public data sources were used to get the information [27]. The distribution of binary and multi-class data sets is depicted in Figs. 1 and 2, respectively. Figure 3 displays an illustration of a chest X-ray.

3.2 Pre-Processing of Images

Images obtained from unprocessed data sources must undergo preprocessing steps [24]. Resizing an image is the first step in the preparation process. The images are reduced to 128×128 pixels because the datasets are of arbitrary size. The next stage after resizing is data shuffling. The dataset must then be divided into Covid-19 and Pneumonia as the final stage.

Table 1 Dataset sources

Reference	Classes	Images	Links
[28–32]	Covid-19	1626	1. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia
	Normal	1802	2. https://www.kaggle.com/datasets/nabeelsajid917/covid-19-x-ray-10000-images
	Pneumonia	1800	3. https://www.kaggle.com/datasets/tarandeeep97/covid19-normal-posteroanteriorpa-xrays
	Total	5228	

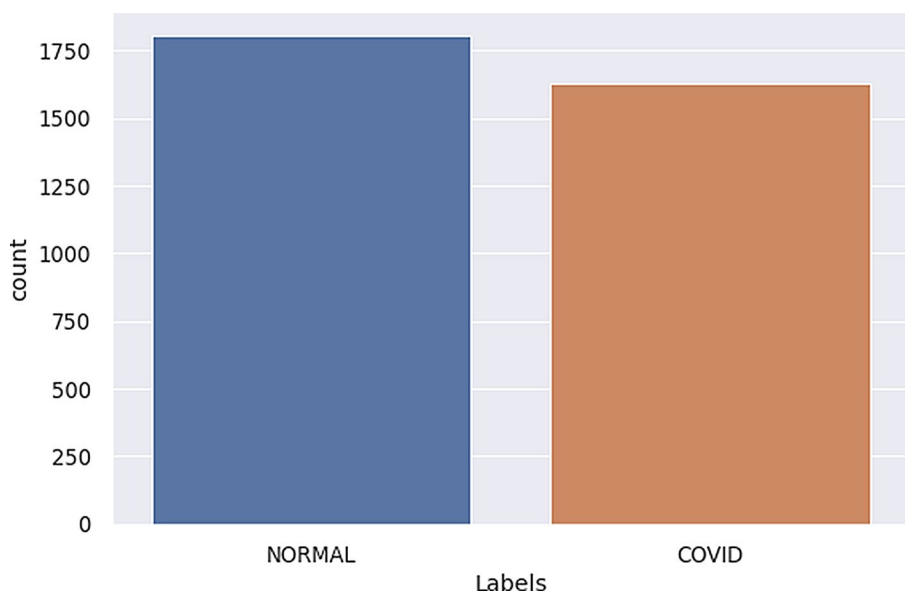


Fig. 1 Binary class data distribution

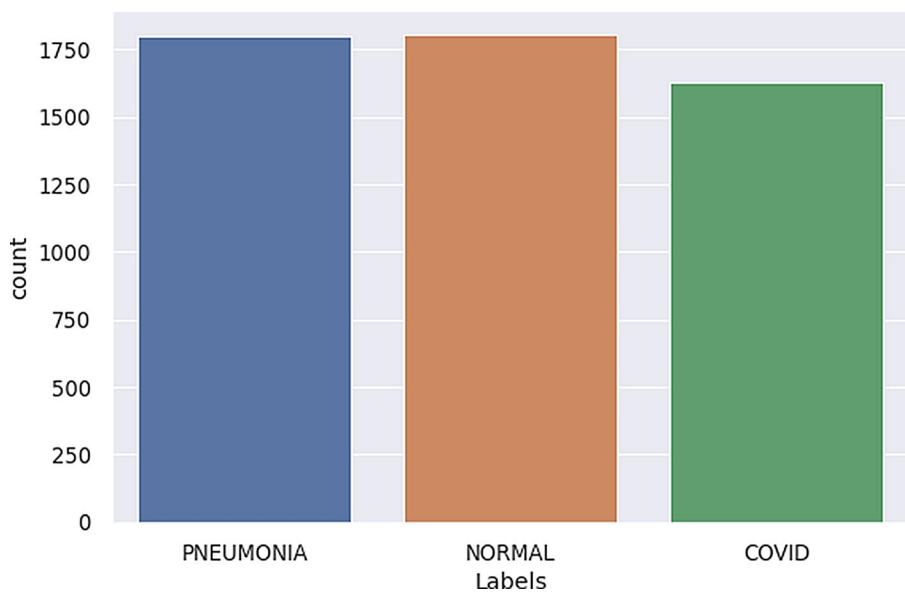


Fig. 2 Multi-class data distribution

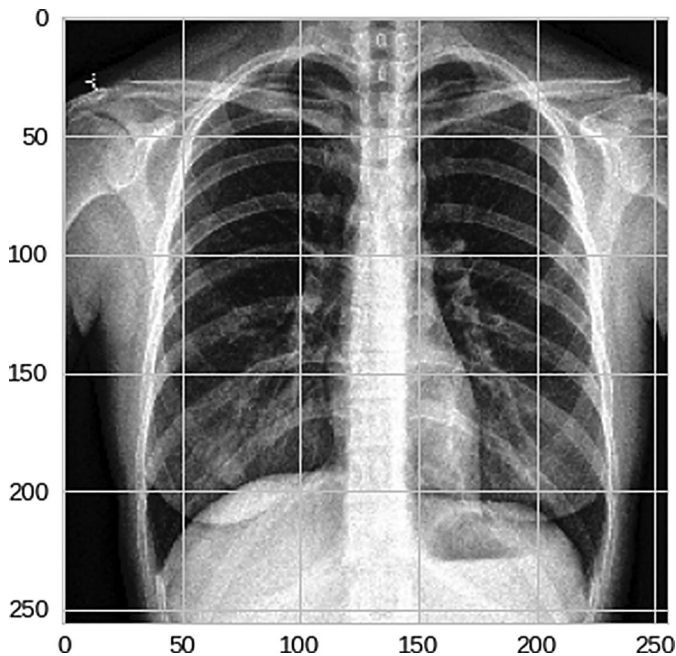


Fig. 3 Sample chest x-ray image

3.2.1 Data Splitting

Training on the complete dataset after the images have been processed is not a viable method. A good strategy is to split the data into two parts where some part of the sample is used for training and another part of the sample is used for testing the model reason being only those samples are used for testing the model which is not seen before. As a result, the dataset is haphazardly distributed into 80% training and 20% testing. This train-test split strategy is employed to evaluate performance and accuracy [33].

3.3 Proposed Classification Framework

Several steps have been followed to design a classification framework. Firstly, all the chest X-ray images are resized, shuffled, and labeled as discussed in Sect. 3.2. The entire dataset is then distributed into train and test, as explained in Sect. 3.3. Training data is utilized to train and construct the suggested model. The final phase is the model's evaluation, done with test data. Figure 4 depicts the categorization framework's overall flow diagram.

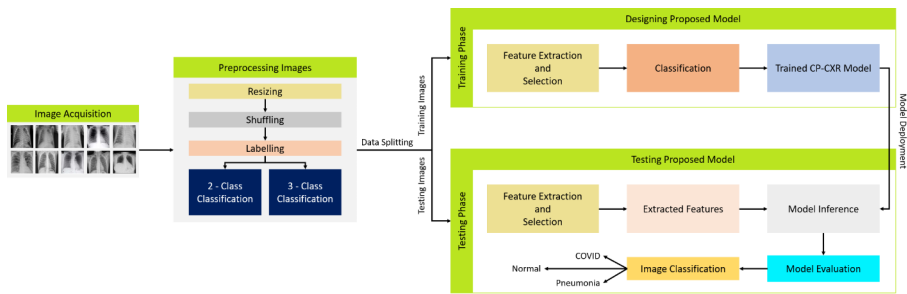


Fig. 4 Proposed classification framework

3.4 Convolutional Neural Network (CNN)

The fundamentals of convolution and related layers employed in our proposed model are explained in this section. Before realizing the Deep CP-CXR model we proposed, we need to have a fundamental knowledge of these concepts.

Input Layer The input layer is the starting point for the deep neural network (DNN) workflow and is responsible for receiving data. In this case, the input is a 128×128 chest X-ray image.

Convolutional Layer The convolutional layer is the foundation for the entire network. It's a deep-learning algorithm that can take images as input. The output is calculated by $Y[i, j]$ over input X at position (i, j) .

$$Y[i, j] = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} X[i + m, j + n] \cdot K[m, n] + b$$

$Y[i, j]$ is the output feature map, $X[i, j]$ is the input image patch, b is the bias term, and $K[m, n]$ is conv kernel of size $k \times k$

Leaky ReLU Layer The network employs a variety of activation functions, resulting in a leaky ReLU layer. In this case, a Leaky Rectified Linear Unit activates the CNN layer (Leaky ReLU). The ReLU activation function has been improved using Leaky ReLU. Here, α is a small constant.

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha x & \text{if } x \leq 0 \end{cases}$$

MaxPooling The MaxPooling layer of the CNN is utilized to downsample the output produced by the preceding layer. This layer is employed to prevent model overfitting. This reduces the size of the image in half in both dimensions. Output of pool size 2×2 is-

$$Y[i, j] = \max(X[2i, 2j], X[2i + 1, 2j], X[2i, 2j + 1], X[2i + 1, 2j + 1])$$

Fully Connected Layer A fully connected layer, also known as the FC or dense layer, is a fully connected layer.

$$Z = W \cdot x + b$$

W is the weight matrix, x is the input vector, b is the bias, and z is the output vector [34].

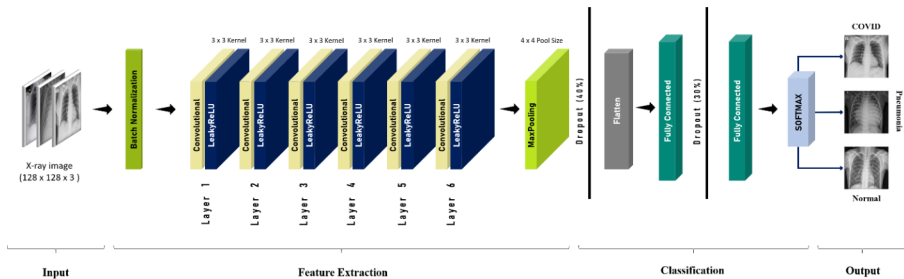
3.5 Proposed Deep CP-CXR Model

In this section, we explain our proposed Deep CP-CXR model for rapid screening purposes so that directed analysis can be accomplished. Deep CP-CXR is a deep learning model that can be used for binary and multi-class classification. Now, when it comes to naming the proposed model, the word “Deep” means a deep neural network, which is utilized to measure the depth of the network. “CP-CXR” stands for Covid and Pneumonia (CP) discovery using chest X-ray (CXR) images.

The proposed Deep CP-CXR model was developed to automate the differentiation between those diagnosed with Covid-19, pneumonia, and those deemed healthy, with the concurrent objectives of minimizing errors and improving efficiency. Rapid discovery of Covid-19 cases permits prompt treatment and seclusion of patients, limiting the spread of the pandemic. Furthermore, after reviewing the literature on the use of chest X-ray images to discover Covid-19, we discovered that the majority of the study used transfer learning approaches. Although a few scholars have suggested their deep learning model for identifying Covid-19, the outcomes in both situations are still ineffective, prompting us to develop our deep learning model. Due to a lack of data on COVID-19 chest X-ray pictures, firstly, the researchers used data improvement methods to lengthen the length of the image. To train our model, we integrated data from various sources, as shown in Table 1. Our Deep CP-CXR model, on the other hand, can cope with the limited accessibility of training chest X-ray pictures and non-uniform data. The proposed Deep CP-CXR model has six convolutional layers, each including a variable filter size and a constant kernel size of 3×3 , followed by the Leaky ReLU activation function. The pool size is 4×4 , and only one global maximum pooling exists. There are two fully connected layers and one flat layer. Table 2 shows the proposed model’s layered description and layer types, their order, output form, number of parameters, and total parameters. There are 4.2 million parameters in total. Figure 5 shows the suggested algorithm and the proposed Deep CP-CXR model.

Table 2 Deep CP-CXR model description

Layer Type	Kernel Size	Pool Size	Output Shape	#Parameters
batch_normalization_20	-	-	(None, 128, 128, 3)	12
conv2d_107 (Conv2D)	(3 × 3)	-	(None, 128, 128, 16)	448
leaky_re_lu_106 (LeakyReLU)	-	-	(None, 128, 128, 16)	0
conv2d_108 (Conv2D)	(3 × 3)	-	(None, 128, 128, 16)	2320
leaky_re_lu_107 (LeakyReLU)	-	-	(None, 128, 128, 16)	0
conv2d_109 (Conv2D)	(3 × 3)	-	(None, 128, 128, 16)	2320
leaky_re_lu_108 (LeakyReLU)	-	-	(None, 128, 128, 16)	0
conv2d_110 (Conv2D)	(3 × 3)	-	(None, 128, 128, 32)	4640
leaky_re_lu_109 (LeakyReLU)	-	-	(None, 128, 128, 32)	0
conv2d_111 (Conv2D)	(3 × 3)	-	(None, 128, 128, 32)	9248
leaky_re_lu_110 (LeakyReLU)	-	-	(None, 128, 128, 32)	0
conv2d_112 (Conv2D)	(3 × 3)	-	(None, 128, 128, 32)	9248
leaky_re_lu_111 (LeakyReLU)	-	-	(None, 128, 128, 32)	0
max_pooling2d_18 (MaxPooling)	-	(4 × 4)	(None, 32, 32, 32)	0
dropout_36 (Dropout)	-	-	(None, 32, 32, 32)	0
flatten_18 (Flatten)	-	-	(None, 32768)	0
dense_36 (Dense)	-	-	(None, 128)	4,194,432
dropout_37 (Dropout)	-	-	(None, 128)	0
dense_37 (Dense)	-	-	(None, 3)	387

**Fig. 5** Proposed deep CP-CXR model

4 Experiment Evaluation and Results

4.1 Implementation Environment

The experiment was performed on a Dell Inspiron-centered implementation environment. A powerful Intel Core i5 5th Generation CPU was installed in this system. It had a generous amount of storage and memory capacity, with 8 gigabytes of random access memory (RAM) and 1 terabyte (TB) of hard disc space. The operating system was Windows 10 in its most basic language configuration, which provided both reliability and simplicity. An NVIDIA GeForce 920 M GPU was added to increase processing power, enabling quicker model training and inference. Python 3.7.0 was used along with PyCharm Community Edition 2021.1.2 as the main software platform. The environment for the implementation made use of crucial libraries includ-

ing Keras, Matplotlib, TensorFlow, NumPy, Pandas, and Scikit, which made it easier to handle data and create and test models. The resources, software, and development tools required for the experiment's successful completion were made available by this carefully designed ecosystem. This setup allowed to perform the correct classification of Covid-19 and pneumonia cases utilizing chest X-ray images, making medical diagnosis and patient care more advanced.

4.2 Parameter Tunning

To prevent overfitting, 32 epochs were used throughout the model's training process, with a batch size of 25. The model training procedure is terminated early when the verification score reaches its maximum and keeps rising. To adjust features like weight to minimize loss and learning rate during compilation, an optimizer is needed. The starting learning rate and momentum are both set to 0.8 for the proposed model, which is compiled to 1e-3 using the adamax optimizer. Cross entropy classification is employed as the loss function.

Mathematically, cross-entropy loss function is derived as below. Where, y_i is true label, $\hat{y}_i$ is predicted probability and N is number of classes.

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

In order to foster the model to converge more quickly and precisely, cross-entropy is preferred because it penalizes confident but wrong predictions strongly. Since it guarantees probabilistic interpretation of the outputs, it is also ideal for multi-class classification and works well with softmax outputs.

4.3 Performance Matrices

Researchers have used a variety of techniques in order to investigate Covid-19 and pneumonia. Accuracy, specificity, sensitivity, and precision are only few of the metrics that are utilized in this approach; other metrics include an F1 score. These metrics offer insightful information about how well detection techniques operate. When the estimation is accurate in projecting as positive, it is said to be true positive (TP). False positives, on the other hand, occur when an observation is negative but the forecast incorrectly predicts a positive outcome. When the observation result is negative and was correctly expected to be negative, the circumstance is referred to as a true negative (TN). Last but not least, a false negative (FN) occurs when an observation is positive but the predictor interprets it as a false negative [3]. The research industry is facilitated in its search for improved diagnostic capacities by these essential principles and measures, which serve as the basis for evaluating and improving the precision and effectiveness of detection approaches used to combat Covid-19 and pneumonia [33].

Accuracy The calculation is executed by dividing the number of accurately identified Covid-19 samples by the total number of Covid-19 samples. In the case of Pneumo-

nia, it is premeditated by distributing the amount of properly classify Pneumonia samples by the entire amount of Pneumonia samples. This is shown in Eq. 1 [35].

$$\text{Accuracy (for each class)} = \frac{TP + TN}{TP + FP + FN + TN} \quad (1)$$

Specificity It is carried out by evenly dividing the total number of True Negative samples by the total number of True Negative and False Positive samples. Equation 2 illustrates this.

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (2)$$

Sensitivity By dividing the total number of True Positive and False Negative samples by the total number of True Positive and False Negative samples, it is determined. Equation 3 depicts it [36].

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (3)$$

Precision Precision is premeditated by distributing the amount of True Positive samples by the quantity of False Positive samples [29]. It is shown in Eq. 4.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

F1-Score It is known as the harmonic mean of Precision and Recall [37]. It is shown in Eq. 5.

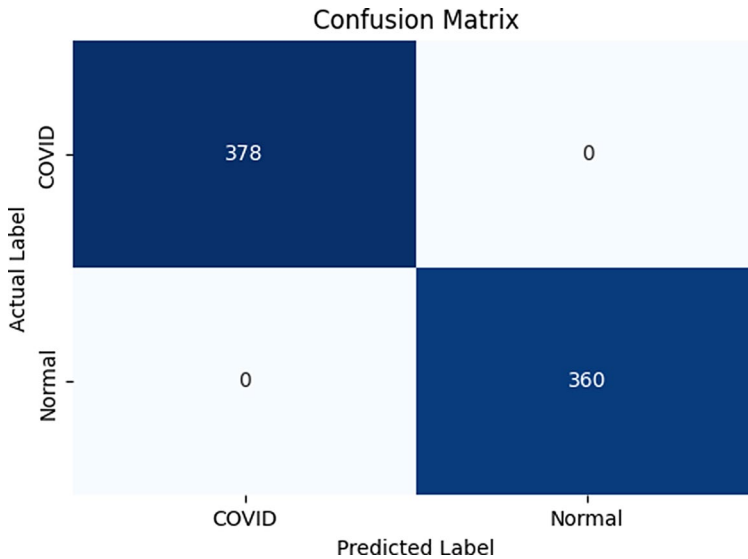
$$\text{F1 - Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

4.4 Experimental Results

Two experiments were carried out. The researchers classified Covid-19 infected individuals and normal patients in the first experiment (binary category). The researchers tested patients diseased with Covid-19, pneumonia patients, and vigorous patients in the second trial (various categories). Performance parameters were utilized to assess the model's effectiveness. The fallouts of the "Deep CP-CXR" projected for binary and multi-class trials are revealed in Table 3. In the two-classification experiment, the model was 100% accurate, while in the multi-classification experiment, the model was 98.57% accurate overall.

Table 3 Binary and multi-class performance matrices of proposed “deep CP-CXR”

Experiment	Precision	Sensitivity	Specificity	F1-Score	Overall Accuracy
Binary Class (Covid-19 and Normal)	100	100	100	100	100%
Multi-Class (Covid-19, Pneu- monia, Normal)	98.00	98.00	100	98.00	98.57%
	97.00	100	100	99.00	
	100	97.00	100	99.00	

**Fig. 6** Binary class confusion matrix

4.4.1 Binary Class Experiment

The results of binary classification are reported in these segments. A total of 2742 photos were used in the training process, with 738 images being used in the testing process. Two far-flung groups, such as Covid-19 and Normal, were examined. Table 3 shows the Covid-19 binary classification performance findings. Similarly, the Covid-19 category’s average accuracy, sensitivity, specificity, and F1 score are all of 100%. The Normal category’s average accuracy, sensitivity, specificity, and F1 score are all 100%. The Covid-19 category and the general category both have a 100% accuracy rate as in Fig. 6 and accuracy loss curves as shown in Fig. 7.

4.4.2 Multi-Class Experiment

Multiple classification results are reported in these sections. A total of 4182 images were used in the training process, with 1045 images being used in the testing process. Covid-19, Normal, and Pneumonia were three distant types studied. The outcome of each. Table 3 shows the multi-class performance results for Covid-19, Normal,

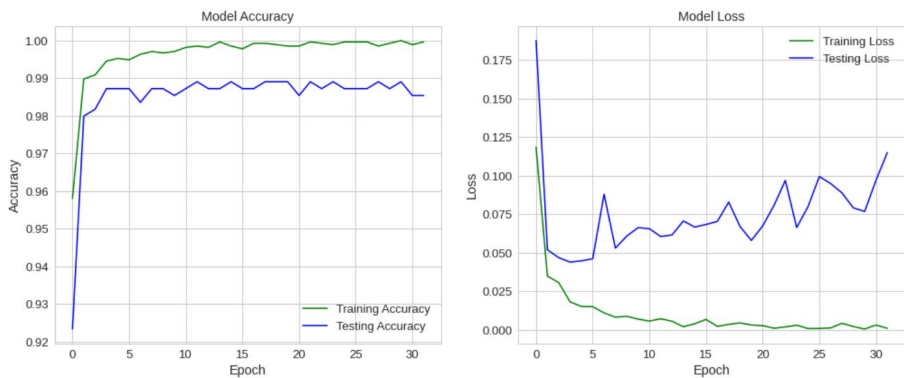


Fig. 7 Model accuracy and loss curve (Binary class)

and Pneumonia. The average accuracy, sensitivity, and F1 score are 98%, while the specificity is 100%. Similarly, the Normal category's average accuracy, specificity, sensitivity, and F1 score are 97%, 100%, 100%, and 99% for the Covid-19 category. Precision, sensitivity, specificity, and F1-score are all at 100% on average. 97%, 100%, and 99%, respectively. The confusion matrices of the multi-class and accuracy loss curves are revealed in Figs. 8 and 9, individually. The heatmap maps obtained by the proposed technique has been shown in Fig. 10.

4.5 Comparative Analysis

Table 4 provides a thorough comparison of the methods currently used to find Covid-19 and pneumonia. For researchers and practitioners aiming to assess and comprehend the numerous approaches used in the field, this table is an invaluable resource. The table identifies the salient features and traits of each strategy and offers details on their advantages, drawbacks, and performance indicators. Researchers can get a comprehensive understanding of the many methods used, such as deep learning models, machine learning algorithms, or hybrid approaches, by looking at this comparative comparison. The table also demonstrates the various data sources, preprocessing approaches, feature extraction strategies, and classification algorithms used by various research. By providing such a broad perspective, this article helps researchers choose the method that best fits their needs.

5 Discussion

The size of the Covid-19 datasets that are currently accessible is modest and inadequate for training. As a result, we use six distinct open-source programs to explore. As indicated in Sect. 4.5, we compared our proposed Deep CP-CXR model with other recently developed models to show the validity and importance of our study effort.

The WHO recommends RT-PCR for Covid-19 detection. However, it is risky to take samples from patients who have COVID-19. Furthermore, due to the RT-PCR

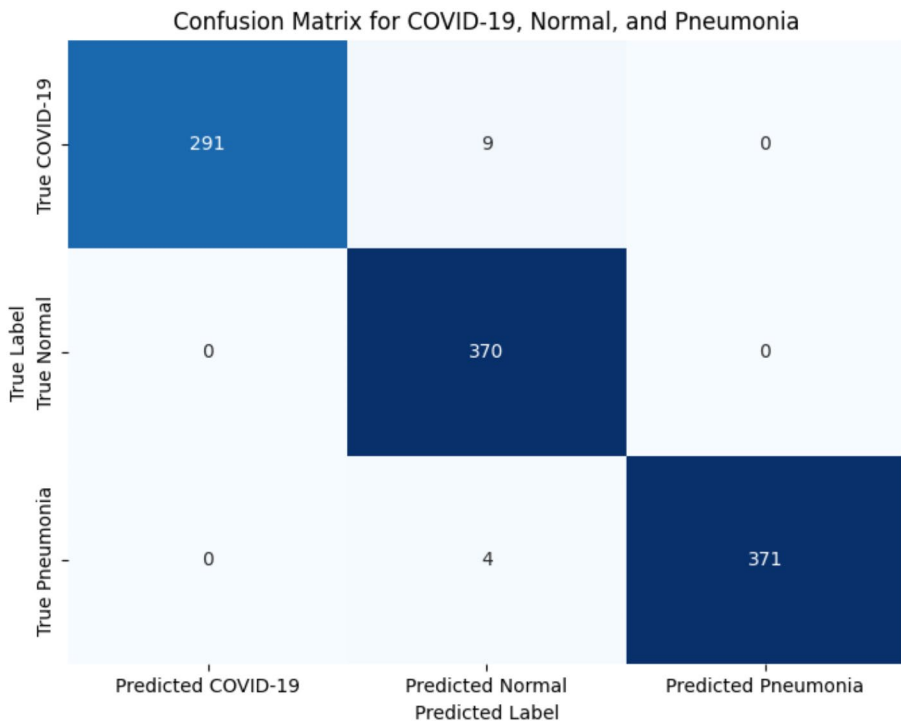


Fig. 8 Multi-class confusion matrix

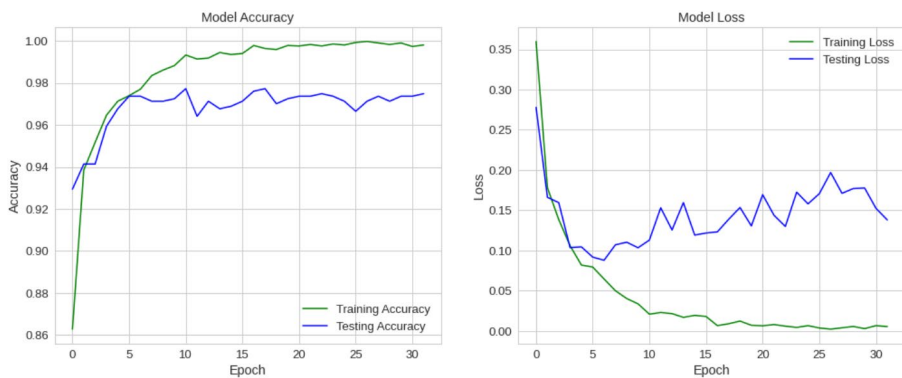
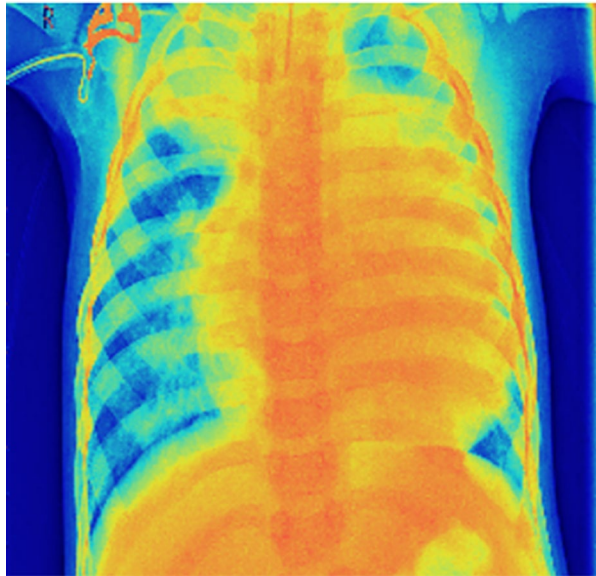


Fig. 9 Model accuracy and loss curve (Multi-class)

test's low accuracy, researchers and medical professionals are driven to develop alternative diagnostic methods. The primary objective of this research project is to provide a framework for clinical decision-making that can identify individuals with pneumonia and COVID-19 early on. Tools are taught using datasets acquired from many sources, and the trained tools may show that diagnosis and early detection of radiologists is Useful. This practice saves time by using chest X-ray pictures to

Fig. 10 Chest x-ray image heat map



find Covid-19 and pneumonia. The model we propose can classify Covid-19 and patients affected by pneumonia without any human intervention, cost-effective and more accurate. Therefore, we believe that this projected model may assist health professionals in the early diagnosis of Covid-19 and pneumonia. Covid-19, an epidemic, is mainly disseminated through interaction with a diseased person when coughing / sneezing. The Covid-19 outburst happened in China and then disseminate to other parts of the world, which has increased the number of deaths. It is a respiratory disease with symptoms such as cough, fever, and difficulty breathing. Many countries have implemented blockade procedures to stop the chain of infection. The government officially announced the blockade measures disrupted the lives of billions of people, and at the same time caused the economic collapse, and even reported that the unemployment rate hit a record high. Therefore, this research calculated the influence of Covid-19 on the tourism industry of affected countries around the world. The results showed that Covid-19 caused panic in the community and increased the demand for tourism. Therefore, as stated in the market equilibrium of the theory of supply and demand, the value of the tourism sector continues to decline as demand decreases [48].

6 Conclusion and Future Work

In this study, we introduced the Deep CP-CXR model, a deep learning architecture leveraging chest X-ray images for precise detection of pneumonia and Covid-19 in patients. Our evaluation comprised two distinct experiments: a binary-class classification (Experiment 1) and a multi-class classification (Experiment 2). A notable level of accuracy was attained in our study, with a 100% accuracy rate observed for binary-

Table 4 Comparative study on Covid-19 and pneumonia detection

Reference	Technique(s)	Classification Type	Dataset Used in the Study	Precision	Sensitivity	Specificity	F1-Score	Accuracy
[38]	DarkCovidNet	Binary class (Covid, No-Findings)	https://github.com/ieee8023/COVID-chestxray-dataset	98.03	95.13	95.30	96.51	98.08
		Multi-class (Covid, No findings, Pneumonia)		89.96	85.35	92.18	87.37	87.02
[39]	CNN model "COVID-ScreenNet"	Multi-class	https://github.com/ieee8023/covid-chestxray-dataset	N/A	N/A	N/A	N/A	97.71
[21]	Discriminative Cost-Sensitive Learning (DCSL)	Multi-class	https://www.kaggle.com/andrewmvd/covid19-x-rays	97	97.09	N/A	96.98	97.01
[40]	Fusion Method	Multi-class	https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia	96.80	99.10	98.20	NA	95.40
[41]	SVM	Binary class	https://github.com/jurader/covid19_xp	N/A	N/A	N/A	N/A	94.12
[42]	VGG-19	Multi-class (Covid, Pneumonia, Normal)	https://github.com/ieee8023/covid-chestxray-dataset	N/A	98.66	96.46	N/A	96.78
[43]	CVDNet	Multi-class (Covid-19, Normal, Pneumonia)	https://www.kaggle.com/tawsifurrahman/covid19-radiography-database	96.72	N/A	N/A	96.68	96.69

Table 4 (continued)

Reference	Technique(s)	Classification Type	Dataset Used in the Study	Precision	Sensitivity	Specificity	F1-Score	Accuracy
[44]	VGG-CapsNet	Binary class (Covid-19, non-Covid-19)	https://doi.org/10.34740/KAGGLE/DSV/1019469	N/A	N/A	N/A	N/A	97
		Multi-class (Covid-19, Normal, Pneumonia)		N/A	N/A	N/A	N/A	92
[45]	CNN	Binary class (Covid-19, non-Covid-19)	1. https://www.kaggle.com/hgunraj/covidxct?select=2A_images 2. https://www.kaggle.com/prashant268/chest-xray-covid19-pneumonia	99.56	99.58	NA	99.59	99.64
		Multi-class (Covid-19, Normal, Pneumonia)		98.22	98.25	NA	98.23	98.28
[46]	Fine-tuned Resnet50	Binary class (Covid-19, non-Covid-19)	https://www.kaggle.com/paultimothymoney/chest-xray-pneumonia	97.00	98.25	97.00	97.00	97.20
	Fine-tuned VGG16	Binary class (Covid-19, Normal)		98.00	98.25	98.33	98.00	98.30
	Fine-tuned InceptionV3	Binary class (Covid-19, non-Covid-19)		98.00	99.25	98.00	98.00	98.10

Table 4 (continued)

Reference	Technique(s)	Classification Type	Dataset Used in the Study	Precision	Sensitivity	Specificity	F1-Score	Accuracy
[46]	ResNet50 + Dense121 + Bi Dir. LSTM	Binary class (Covid-19, Normal Cases)	https://lhncbc.nlm.nih.gov/publication/pub9931	98.22	N/A	N/A	94.22	98.22
[47]	SWT + iChi2 + SVM	Multi-class (Covid-19, Normal, Pneumonia)	https://www.kaggle.com/tawsifurrahman/covid19-radiography-database	99.241	N/A	N/A	99.241	99.24
Proposed Deep CP-CXR		Binary class (Covid-19, Normal)	Six distinct public data sources	100	100	100	100	100
		Multi-class (Covid-19, Normal, Pneumonia)		98.33	98.33	100	98.67	98.57

*N/A: Not Available

class classification and a remarkable 98.57% accuracy rate reached for the multi-class classification task that is in identifying between Covid-19 and pneumonia.

Our findings mark a significant stride in the advancement of automated medical diagnostic applications, leveraging state-of-the-art multimodal deep learning models. The Deep CP-CXR model, in conjunction with these approaches, showcases the potential to reshape current medical diagnostic workflows. By effectively examining an enormous quantity of X-ray images and digital CT scan data, these advancements have the potential to ease the workload and time restrictions faced by radiologists. The potential implications of the Deep CP-CXR model and its corresponding techniques on medical diagnostics are considerable. The utilization of automated systems for the swift and precise detection of diseases, such as pneumonia and Covid-19, expedites the diagnostic procedures and empowers healthcare practitioners to make prompt and knowledgeable decisions. The integration of cutting-edge technologies like deep learning holds promise for enhancing medical professionals' knowledge and serving as a valuable tool in critical clinical scenarios where swift and precise disease identification is imperative.

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Data Availability Publicly Available. <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>. <https://www.kaggle.com/datasets/nabeelsajid917/covid-19-x-ray-10000-images>. <https://www.kaggle.com/datasets/tarandeep97/covid19-normal-posteroanteriorpa-xrays>.

Declarations

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Competing Interest Not Applicable.

References

1. Zhu H, Wei L, Niu P (2020) The novel coronavirus outbreak in Wuhan, China. *Glob Heal Res Policy* 5(1):1–3. <https://doi.org/10.1186/S41256-020-00135-6/METRICS>
2. Zhang H, Du F, Cao X jun, et al (2021) Clinical characteristics of coronavirus disease 2019 (COVID-19) in patients out of Wuhan from China: a case control study. *BMC Infect Dis* 21(1):1–7. <https://doi.org/10.1186/s12879-021-05897-z>
3. Shibly KH, Dey SK, Islam MTU, Rahman MM (2020) COVID faster R–CNN: A novel framework to diagnose novel coronavirus disease (COVID-19) in X-Ray images. *Inf Med Unlocked* 20. <https://doi.org/10.1016/j.imu.2020.100405>
4. Fehr AR, Perlman S (2015) Coronaviruses: An overview of their replication and pathogenesis. In: *Coronaviruses: Methods and Protocols*. Vol 1282. *Methods Mol Biol* 1–23. https://doi.org/10.1007/978-1-4939-2438-7_1
5. Punns NS, Sonbhadra SK, Agarwal S (2020) COVID-19 Epidemic Analysis using Machine Learning and Deep Learning Algorithms. *medRxiv*. 20057679. <https://doi.org/10.1101/2020.04.08.20057679>
6. Ndwanwe D, Wiysonge CS (2021) COVID-19 vaccines. *Curr Opin Immunol* 71:111–116. <https://doi.org/10.1016/j.coi.2021.07.003>
7. Varshni D, Thakral K, Agarwal L, Nijhawan R, Mittal A (2019) Pneumonia detection using CNN based feature extraction. *Proc 2019 3. IEEE Int Conf Electr Comput Commun Technol ICECCT 2019*. <https://doi.org/10.1109/ICECCT.2019.8869364>
8. Hashmi MF, Katiyar S, Keskar AG, Bokde ND, Geem ZW (2020) Efficient pneumonia detection in chest Xray images using deep transfer learning. *Diagnostics* 10(6). <https://doi.org/10.3390/DIAGNOSTICS10060417>
9. Khalifa NEM, Smarandache F, Manogaran G, Loey M (2021) A study of the neutrosophic set significance on deep transfer learning models: an experimental case on a limited COVID-19 chest X-ray dataset. <https://doi.org/10.1007/S12559-020-09802-9>. *Cognit Comput doi*
10. Ucar F, Korkmaz D (2020) COVIDiagnosis-Net: deep Bayes-SqueezeNet based diagnosis of the coronavirus disease 2019 (COVID-19) from X-ray images. *Med Hypotheses* 140:109761. <https://doi.org/10.1016/j.mehy.2020.109761>

11. Behzadi-khormouji H, Rostami H, Salehi S et al (2019) Deep learning, reusable and problem-based architectures for detection of consolidation on chest X-ray images. *Comput Methods Programs Biomed* 185:105162–105162. <https://doi.org/10.1016/j.CMPB.2019.105162>
12. Yi R, Tang L, Tian Y, Liu J, Wu Z (2021) Identification and classification of pneumonia disease using a deep learning-based intelligent computational framework. *Neural Comput Appl* 1–14. <https://doi.org/10.1007/S00521-021-06102-7/TABLES/5>
13. Purohit K, Kesarwani A, Ranjan Kisku D, Dalui M (2022) COVID-19 detection on chest X-Ray and CT scan images using Multi-image augmented. *Deep Learn Model* 395–413. https://doi.org/10.1007/978-981-16-6890-6_30
14. Johri S, Goyal M, Jain S, Baranwal M, Kumar V, Upadhyay R (2021) A novel machine learning-based analytical framework for automatic detection of COVID-19 using chest X-ray images. *Int J Imaging Syst Technol* 31(3):1105–1119. <https://doi.org/10.1002/IMA.22613>
15. Zhang YD, Satapathy SC, Liu S, Li GR (2021) A five-layer deep convolutional neural network with stochastic pooling for chest CT-based COVID-19 diagnosis. *Mach Vis Appl* 32(1). <https://doi.org/10.1007/S00138-020-01128-8>
16. Gozes O, Frid-Adar M, Greenspan H et al (2020) Rapid AI Development Cycle for the Coronavirus (COVID-19) Pandemic: Initial Results for Automated Detection & Patient Monitoring using Deep Learning CT Image Analysis. *arXiv*. Accessed January 21, 2025. <https://arxiv.org/abs/2003.05037v3>
17. Altaf F, Islam SMS, Janjua NK (2021) A novel augmented deep transfer learning for classification of COVID-19 and other thoracic diseases from X-rays. *Neural Comput Appl* 33(20):14037–14048. <https://doi.org/10.1007/S00521-021-06044-0/TABLES/7>
18. Rangarajan AK, Ramachandran HK (2021) A preliminary analysis of AI based smartphone application for diagnosis of COVID-19 using chest X-ray images. *Expert Syst Appl* 183:115401. <https://doi.org/10.1016/j.ESWA.2021.115401>
19. McPhedran R, Toombs B (2021) Efficacy or delivery? An online discrete choice experiment to explore preferences for COVID-19 vaccines in the UK. *Econ Lett* 200. <https://doi.org/10.1016/J.ECONLET.2021.109747>
20. Mehta N, Sales RM, Babagbemi K et al (2021) Unilateral axillary adenopathy in the setting of COVID-19 vaccine: Follow-up. *Clin Imaging* 80:83. <https://doi.org/10.1016/J.CLINIMAG.2021.06.037>
21. Li T, Han Z, Wei B, Zheng Y, Hong Y, Cong J (2020) Robust Screening of COVID-19 from Chest X-ray via Discriminative Cost-Sensitive Learning. *arXiv* Accessed January 21, 2025. <https://arxiv.org/abs/2004.12592v2>
22. Hertel R, Benlamri R (2021) COV-SNET: A deep learning model for X-ray-based COVID-19 classification. *Inf Med Unlocked* 24. <https://doi.org/10.1016/J.IMU.2021.100620>
23. Alorf A (2021) The practicality of deep learning algorithms in COVID-19 detection: application to chest X-ray images. *Algorithms* 14(6). <https://doi.org/10.3390/A14060183>
24. Shastri S, Singh K, Deswal M, Kumar S, Mansotra V (2021) CoBiD-net: a tailored deep learning ensemble model for time series forecasting of covid-19. *Spat Inf Res* 1–14. <https://doi.org/10.1007/s41324-021-00408-3>
25. Sitaula C, Aryal S (2021) New bag of deep visual words based features to classify chest x-ray images for COVID-19 diagnosis. *Heal Inf Sci Syst* 9(1):1–12. <https://doi.org/10.1007/S13755-021-00152-W/TABLES/5>
26. Wang W, Li Y, Li J, Zhang P, Wang X (2021) Detecting COVID-19 in chest X-Ray images via MCFNet. *Comput Intell Neurosci* 2021. <https://doi.org/10.1155/2021/3604900>
27. Rahman T, Khandakar A, Qiblawey Y et al (2021) Exploring the effect of image enhancement techniques on COVID-19 detection using chest X-ray images. *Comput Biol Med* 132:104319. <https://doi.org/10.1016/j.combiomed.2021.104319>
28. Abbas A, Abdelsamea MM, Gaber MM (2021) Classification of COVID-19 in chest X-ray images using detrac deep convolutional neural network. *Appl Intell* 51(2):854–864. <https://doi.org/10.1007/s10489-020-01829-7>
29. Chest X-R, Images (Pneumonia) (eds) (2025) Chest X-Ray Images (Pneumonia), Kaggle Accessed May 02, 2018 [Online] Available. Published online 2018:1–1. <https://www.kaggle.com/datasets/paul-timothymooney/chest-xray-pneumonia> Accessed January 17
30. COVID-19 patients lungs X-ray images 10000 (2025) Google Sch. Published online 2020. <https://www.kaggle.com/datasets/nabeelsajid917/covid-19-x-ray-10000-images> Accessed January 17
31. COVID-19 Image Data Collection *arXiv*. [https://doi.org/10.1016/S0140-6736\(20\)30211-7](https://doi.org/10.1016/S0140-6736(20)30211-7)

32. Singh T (2020) COVID-19 & Normal Posteroanterior (PA) X-rays. Kaggle. Published <https://www.kaggle.com/datasets/tarandeep97/covid19-normal-posteroanteriorpa-xrays> Accessed January 17, 2025
33. Kumar S, Shastri S, Mahajan S et al (2022) LiteCovidNet: A lightweight deep neural network model for detection of COVID-19 using X-ray images. *Int J Imaging Syst Technol* 32(5):1464–1480. <https://doi.org/10.1002/ima.22770>
34. Ayalew AM, Salau AO, Tamyalew Y, Abeje BT, Woreta N (2023) X-Ray image-based COVID-19 detection using deep learning. *Multimed Tools Appl* 82(28):44507–44525. <https://doi.org/10.1007/s11042-023-15389-8>
35. Kwon N, Lee D, Yoon YH, Youn I, Moon HJ, Han S (2025) COVID-19 and flu diagnosis from short electrocardiogram images using a residual neural network. *Biomed Signal Process Control* 103:107408. <https://doi.org/10.1016/j.bspc.2024.107408>
36. Shastri S, Kansal I, Kumar S, Singh K, Popli R, Mansotra V (2022) CheXImageNet: a novel architecture for accurate classification of Covid-19 with chest x-ray digital images using deep convolutional neural networks. *Health Technol (Berl)* 12(1):193–204. <https://doi.org/10.1007/s12553-021-00630-x>
37. Jain PK, Quamer W, Pamula R (2021) Sports result prediction using data mining techniques in comparison with base line model. *OPSEARCH* 58(1):54–70. <https://doi.org/10.1007/s12597-020-00470-9>
38. Ozturk T, Talo M, Yildirim EA, Baloglu UB, Yildirim O, Rajendra Acharya U (2020) Automated detection of COVID-19 cases using deep neural networks with X-ray images. *Comput Biol Med* 121(April):103792. <https://doi.org/10.1016/j.combiomed.2020.103792>
39. Dhaka VS, Rani G, Oza MG, Sharma T, Misra A (2021) A deep learning model for mass screening of COVID-19. *Int J Imaging Syst Technol* 31(2):483–498. <https://doi.org/10.1002/IMA.22544>
40. Shorfuzzaman M, Masud M, Alhumyani H, Anand D, Singh A (2021) Artificial neural Network-Based deep learning model for COVID-19 patient detection using X-Ray chest images. *J Healthc Eng* 2021. <https://doi.org/10.1155/2021/5513679>
41. Khan MA (2021) An automated and fast system to identify COVID-19 from X-ray radiograph of the chest using image processing and machine learning. *Int J Imaging Syst Technol* 31(2):499–508. <https://doi.org/10.1002/IMA.22564>
42. Apostolopoulos ID, Mpesiana TA (2020) Covid-19: automatic detection from X-ray images utilizing transfer learning with convolutional neural networks. *Phys Eng Sci Med* 43(2):635–640. <https://doi.org/10.1007/s13246-020-00865-4>
43. Ouchicha C, Ammor O, Meknassi M (2020) CVDNet: A novel deep learning architecture for detection of coronavirus (Covid-19) from chest x-ray images. *Chaos Solitons Fractals* 140:110245. <https://doi.org/10.1016/J.CHAOS.2020.110245>
44. Tiwari S, Jain A (2021) Convolutional capsule network for COVID-19 detection using radiography images. *Int J Imaging Syst Technol* 31(2):525–539. <https://doi.org/10.1002/IMA.22566>
45. Thakur S, Kumar A (2021) X-ray and CT-scan-based automated detection and classification of covid-19 using convolutional neural networks (CNN). *Biomed Signal Process Control* 69:102920. <https://doi.org/10.1016/J.BSPC.2021.102920>
46. Rasheed J, Alsubai S (2023) A hybrid deep fused learning approach to segregate infectious diseases. *Comput Mater Contin* 74(2):4239–4259. <https://doi.org/10.32604/cmc.2023.031969>
47. Rasheed J, Shubair RM (2022) Screening lung diseases using cascaded feature generation and selection strategies. *Healthc* 10(7):1313. <https://doi.org/10.3390/healthcare10071313>
48. Bakar NA, Rosbi S (2020) Effect of coronavirus disease (COVID-19) to tourism industry. *Int J Adv Eng Res Sci* 7(4):189–193. <https://doi.org/10.22161/ijaers.74.23>

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